Demand totals and daily profile

M codes refer to the shared metric contract in the requirements document and dictionary.

Q01. How does monthly demand energy change across the five operating areas?

Plan for recurring changes in electricity needs.

Mapping	Specification
Facts	FactHourlyDemand
Dimensions	DimDate, DimBalancingAuthority
Measures	M01
Actual source fields	Demand (MW), Data Date, Balancing Authority
Expected output	Monthly MWh by authority, with eligible and missing hours.
Dependency / limit	Null actual demand reduces measured coverage. Include leap day and local operating dates.

Q02. Which local hours have the highest average demand?

Identify the normal daily demand profile for each operating area.

Mapping	Specification
Facts	FactHourlyDemand
Dimensions	DimHour, DimDate, DimBalancingAuthority
Measures	M02
Actual source fields	Demand (MW), Local Time at End of Hour, Data Date, Balancing Authority
Expected output	Hour-ending profile in MW, with observation counts.
Dependency / limit	Autumn repeated clock hours are separate observations in the same local-hour bucket.

Peak hours and seasonal patterns

M codes refer to the shared metric contract in the requirements document and dictionary.

Q03. When did each authority reach its monthly peak demand?

Identify observed periods requiring the greatest supply response.

Mapping	Specification
Facts	FactHourlyDemand
Dimensions	DimDate, DimHour, DimBalancingAuthority
Measures	M03
Actual source fields	Demand (MW), UTC Time at End of Hour, Local Time at End of Hour, Data Date
Expected output	Peak MW and tied UTC/local timestamps by authority and month.
Dependency / limit	This is an observed hourly-average peak. It does not measure instantaneous system peaks.

Q04. How do weekday and weekend demand patterns differ across seasons?

Compare recurring operating-day patterns within the same area.

Mapping	Specification
Facts	FactHourlyDemand
Dimensions	DimDate, DimBalancingAuthority
Measures	M02
Actual source fields	Demand (MW), Data Date, Balancing Authority
Expected output	Mean MW by authority, calendar year, meteorological season and day type.
Dependency / limit	Weekday/weekend comes from the operating date. Holidays are not inferred.

Forecast accuracy and bias

M codes refer to the shared metric contract in the requirements document and dictionary.

Q05. Which operating areas and months have the largest forecast errors?

Compare day-ahead forecast accuracy using both scale-dependent and normalized measures.

Mapping	Specification
Facts	FactHourlyDemand
Dimensions	DimDate, DimBalancingAuthority
Measures	M04, M05, M06
Actual source fields	Demand (MW), Demand Forecast (MW), UTC Time at End of Hour, Balancing Authority
Expected output	MAE, WAPE, bias and pair count by authority/month.
Dependency / limit	Exclude missing pairs and nonpositive actuals. Published historical forecasts may already contain corrections.

Q06. When do forecasts systematically fall below actual demand?

Detect recurring underforecast patterns for further review.

Mapping	Specification
Facts	FactHourlyDemand
Dimensions	DimHour, DimDate, DimBalancingAuthority
Measures	M04, M07
Actual source fields	Demand (MW), Demand Forecast (MW), local end time and operating date
Expected output	Bias and underforecast percentage by hour, season and authority.
Dependency / limit	Use one matched pair per authority/UTC hour. The project evaluates the publisher forecast; it does not train a forecasting model.

Generation and wind/solar share

M codes refer to the shared metric contract in the requirements document and dictionary.

Q07. How does reported generation by energy source change each month?

Explain the recorded supply mix across areas and periods.

Mapping	Specification
Facts	FactHourlyGeneration
Dimensions	DimEnergySource, DimDate, DimBalancingAuthority
Measures	M08
Actual source fields	All reported Net Generation (MW) from ... columns, Data Date, Balancing Authority
Expected output	Signed MWh by original source and approved roll-up group.
Dependency / limit	Respect the source-header change. Preserve combined legacy hydro/storage and indicate categories without observations.

Q08. What share of reported net generation comes from wind and solar categories?

Compare recorded wind/solar contributions within supported reporting periods.

Mapping	Specification
Facts	FactHourlyGeneration and FactHourlyDemand
Dimensions	DimDate, DimBalancingAuthority, DimEnergySource
Measures	M09, M12
Actual source fields	Reported wind/solar fuel columns, Net Generation (MW), authority and UTC end time
Expected output	Matched-period percentage with eligible/excluded hour counts and source definitions.
Dependency / limit	Integrated battery categories remain labeled. The measure is a reported category share, with signed net values and reconciliation screening.

High demand and source quality

M codes refer to the shared metric contract in the requirements document and dictionary.

Q09. Does the generation mix differ between high-demand and other hours?

Study which sources are recorded during the largest observed loads.

Mapping	Specification
Facts	FactHourlyDemand and FactHourlyGeneration
Dimensions	DimDate, DimBalancingAuthority, DimEnergySource
Measures	M10, M08
Actual source fields	Demand (MW), reported fuel columns, authority and UTC end time
Expected output	Generation mix comparison for demand at/above the monthly 90th percentile versus remaining eligible hours.
Dependency / limit	First determine the demand set, then join aggregated generation. An association does not establish a dispatch decision or causal effect.

Q10. Where do missing measures and generation discrepancies affect reporting?

Prioritize source-quality review and qualify the analytical outputs.

Mapping	Specification
Facts	FactHourlyDemand, FactHourlyGeneration and expected-hour grid
Dimensions	DimDate, DimBalancingAuthority, DimSourceSnapshot
Measures	M11, M12
Actual source fields	Core measures, reported fuel columns, UTC end time, source filename/hash and row number
Expected output	Completeness, invalid/zero/negative counts and reconciliation flags by authority/month.
Dependency / limit	Blank fuel values and structurally absent columns need separate interpretation. Diagnostic flags do not automatically authorize corrections.